

REPLAYABLE MINIMUM-DISTANCE CERTIFICATES FOR STABILIZER CODES, WITH NO SOLVER AND NO PROOF ASSISTANT IN THE TRUSTED BASE: THE BIVARIATE-BICYCLE FAMILY, AND THE EXACT DISTANCE OF $[[288, 12, 18]]$

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Computation and authorship. All searches, encodings, symmetry-breaking constructions, and certificate designs in this work were produced by **Claude Fable 5** (Anthropic), directed by the author, on a single Apple M4 laptop (10 cores, 16 GB). The three independent checkers share no code with the generating pipeline, import only the Python standard library, and were exercised against the corpus by a separate agent instance that was given no access to the pipeline. This is a factual methods statement, and it is part of the point of the paper: the artifacts are designed so that the provenance of the *search* is irrelevant to the validity of the *result*. External tools used by the pipeline only: CaDiCaL 3.0.1 (`--lrat --no-binary`), Python 3.

Prior-art record. The credit statements in §1.4 are the output of a full-text adversarial sweep completed on August 4, 2026, in which every source was read in raw form (arXiv LaTeX source, GitHub repository trees at a named commit, package documentation) and in which two claims of an earlier internal draft were found to be wrong and withdrawn. The dated record of that sweep, listing what was read and what changed, ships with the artifacts as `SWEEP-RECORD-QEC-2026-08-04.md`. The August 6 extension that determines $d([[288, 12, 18]]) = 18$ and adds the $[[360, 12, \leq 24]]$ lower bound rests on a further full-text pass over its own new sources, recorded inline in §6 and §1.4. *Draft of August 6, 2026.*

ABSTRACT. This is a verification contribution. The exact minimum distances of the codes treated here are, with a single exception noted below, already in the literature; what did not exist is a standalone artifact that a third party can replay without a SAT solver and without a proof assistant. We supply one for eleven stabilizer codes, together with three short checkers (a fourth handles the exact-distance ladder below), and we submit the result to an independent audit: 47 certificate checks run, 47 passed, 0 failed; 5.08 GiB of LRAT replayed in pure Python inside a 79 MB peak resident set; all five bivariate-bicycle parity-check matrices rebuilt *byte-identically* from the published construction of Bravyi et al.; 182 of 182 manifest SHA-256 entries matching; 11 of 11 negative controls correctly rejected.

The formats are deliberately small. Upper bounds are witness pairs verified by three $\mathbb{F}_2$ matrix–vector products. Lower bounds are LRAT unsatisfiability proofs for a canonical CNF encoding of “some nontrivial logical has weight at most K ”, in which the checker regenerates the CNF from the raw parity-check matrices and machine-checks the three algebraic side conditions that make the encoding *exact*; neither the solver nor the shipped CNF is trusted, and for every code in the corpus, including the largest proof, replay needs nothing beyond the Python standard library.

For the gross code $[[144, 12, 12]]$ of Bravyi et al. we certify $d = 12$ end to end: weight-12 X - and Z -witnesses, and in each sector a single *symmetry-free* LRAT proof (868 MB for X , 672 MB for Z) of $d_X, d_Z \geq 12$, so that no symmetry lemma enters the trusted base for this code. For $[[288, 12, 18]]$ we determine the distance exactly: $d = 18$, certified end to end and, as far as our sweep could determine, for the first time in a form a third party can replay. The upper bound is a weight-18 witness; the lower bound $d_X \geq 18$ is a two-rung ladder in a profile-normalisation encoding, closed by two on-paper lemmas whose hypotheses are machine-checked — Lemma P, that every X -logical has even weight, and Lemma S, the completeness of the reduction — with the passage to d supplied by the shipped duality certificate. This *confirms* the value Bravyi et al. assert for that code by integer programming, without a checkable artifact; it does not correct it. The strongest lower bound previously *reported* by a certifying-capable method was $d \geq 11$, solver-asserted by Chen, Jafari and Lai with no proof files in their repository and with every configuration timing out. One dependency we flag rather than bury: the exclusion of a weight-16 logical — the rung that reaches 18 rather than 16 — rests on the profile-normalisation encoding alone; an independent, differently structured encoding corroborates the ladder only to $d_X \geq 12$. The same construction yields the first lower bound of any kind for $[[360, 12, \leq 24]]$: we certify $16 \leq d \leq 24$, the lower end by this method and the upper end cited from Bravyi et al.

We state the trusted base explicitly, separating what is machine-checked per certificate from the facts that are assumed, and we report the four defects the audit found, including one latent soundness hole in a checker branch. We claim priority neither for the distance values, which are Bravyi et al.’s, nor for machine-checked quantum distance proofs, which are LEAN-QEC’s and whose public repository reports a completed gross-code verification as of 2026-07-10; and we claim no novelty for the profile-normalisation encoding against automated symmetry-breaking tools, which we did not benchmark. What is offered here is a certificate format and a trusted base — four standard-library readers, 1,128 lines in total, with no solver and no proof assistant in them — the first independently-replayable determination that $d([[288, 12, 18]]) = 18$, the first lower bound of any kind for $[[360, 12, \leq 24]]$, and artifacts that outlive the tools that produced them.

1. INTRODUCTION

1.1. The problem. The minimum distance d of a quantum error-correcting code is the single number that determines how many errors it can detect. It is also expensive: computing it is a minimum-weight problem over a coset space, intractable in general [Var97], and the standard tools for it in the quantum literature are of two kinds. Randomized searches such as QDistRnd [PSKK22] and Stim’s `search_for_undetactable_logical_errors` [Gid21] return

upper bounds and say so plainly; Stim’s documentation is explicit that “THIS IS A HEURISTIC METHOD”. Exact methods — integer programming [LAR11], SAT and MaxSAT [CJL26], branch and bound — return a number and an implicit assertion that the search was exhaustive. The survey of Webster, Jacob and Higgott [WJH26] classifies the landscape carefully; what almost none of it produces is an artifact a third party can check.

This matters more than usual here, because the published distances of the codes now being built into hardware are load-bearing engineering constants. The bivariate-bicycle (BB) codes of Bravyi, Cross, Gambetta, Maslov, Rall and Yoder [BCGMRY] — in particular the $[[144, 12, 12]]$ “gross code” — are the basis of a fault-tolerant memory architecture. Their distances were computed by the mixed-integer-programming method of Landahl, Anderson and Rice [LAR11]; the paper is explicit about this and about which entries of its Table 3 are upper bounds only. The gross-code value has since been confirmed independently, again by mathematical programming: Cruz-Benito, Cross, Kremer and Faro [CCKF26] close the MILP to a gap of zero and obtain $d = 12$ exactly. Nothing is wrong with either computation. But a reader who wants to check them must re-run a solver and trust the re-run; neither emits an object that survives the solver.

1.2. What this note provides. An artifact-first alternative. For each code we ship files that a skeptic can replay in isolation, and three short Python programs that replay them (a fourth, `check_prof.py`, checks the profile-normalisation certificates of §6). The design constraints were:

- (i) *No solver in the trusted base.* The checker never runs a SAT solver and never reads the CNF file the solver saw. It *regenerates* the CNF from the raw parity-check matrices, then replays the LRAT proof against its own clause list. A corrupted shipped CNF changes nothing — this was tested (§7).
- (ii) *No proof assistant in the trusted base.* No Lean, no Mathlib, no toolchain fetch. The reader needs CPython and the files.
- (iii) *The encoding’s exactness is machine-checked, not asserted.* The reduction from “ $d_X \geq K + 1$ ” to “this CNF is unsatisfiable” is valid only under three algebraic side conditions on (H_X, H_Z, P) . The checker verifies all three, per certificate, over the raw matrices (Theorem 3.1).
- (iv) *Everything the checker assumes is written down.* Section 4 is the honest inventory, including the items argued in prose and not machine-verified anywhere.

1.3. Results. Table 1 is the corpus. Every entry traces to a file in `certificates/`; the byte counts are the actual proof sizes and the replay times are wall clock from the independent audit of §7, measured with `/usr/bin/python3` (CPython 3.9.6, no `numpy`, no compiled helper of any kind).

Three things in that table deserve to be stated precisely, because each of them touches published work.

(A) The gross code. We certify $d([[144, 12, 12]]) = 12$ end to end. The upper bound is a pair of weight-12 witnesses, 970-byte JSON files, checked in 7 ms each. The lower bound is certified twice in each direction: once by a symmetry-broken pair of instances in the X sector, and once, in *each* sector, by a single LRAT proof with no symmetry hypothesis at all — 867,803,294 bytes for X and 671,988,205 bytes for Z . The symmetry-free pair is the one that matters for the trusted base: it removes the orbit lemma of §3.3 from the argument entirely, so $d([[144, 12, 12]]) = 12$ rests on nothing but the exactness theorem, the LRAT semantics, and the standard cardinality and XOR encodings.

code	n, k	certified	lower-bd. proof (bytes)	solver (s)	pure-Python replay (s)
Steane $[[7, 1, 3]]$	7, 1	$d = 3$	699 699	0.02 0.07	0.0003 0.0009
five-qubit $[[5, 1, 3]]$	5, 1	$d = 3$	1,514	0.02	0.0004
rot. surface $d=3$	9, 1	$d = 3$	518 746	0.03 0.02	0.0004 0.0004
rot. surface $d=5$	25, 1	$d = 5$	4,685 7,085	0.02 0.02	0.001 0.001
rot. surface $d=7$	49, 1	$d = 7$	24,939 75,107	0.02 0.02	0.005 0.009
Golay $[[23, 1, 7]]$	23, 1	$d = 7$	226,377 each	0.04 0.04	0.024 0.024
BB $[[72, 12, 6]]$	72, 12	$d = 6$	1.54 1.93 MB	0.26 0.30	0.14 0.17
BB $[[90, 8, 10]]$	90, 8	$d = 10$	128 151 MB	48.4 50.8	10.9 13.0
BB $[[108, 8, 10]]$	108, 8	$d = 10$	71.6 82.3 MB	28.2 24.7	6.2 7.1
BB $[[144, 12, 12]]$	144, 12	$d = 12$	868 672 MB [†]	342 227	176 72.9
BB $[[288, 12, 18]]$	288, 12	$d = 18$	358 MB [‡]	2,526	1,649
BB $[[360, 12, \leq 24]]$	360, 12	$16 \leq d \leq 24$	43 MB [§]	231	40

TABLE 1. The certificate corpus. Paired entries are X | Z sector, except the two **d**-in-boldface rows at $n = 288, 360$, whose proofs are in the X sector and are converted to d by duality. Solver times are CaDiCaL 3.0.1 on one Apple M4 laptop, recorded in each code’s `meta.json`. Replay times are from the independent audit (§7), except at $n = 288, 360$, whose profile-normalisation certificates postdate the audit and were replayed separately by `check_prof.py` on a loaded machine; all use nothing outside the Python standard library. [†]Symmetry-free single-instance proofs; a symmetry-broken X certificate (124 MB + 34 kB, solver 45.0 s) is also shipped and replays in 10.1 s. [‡]The exact value $d = 18$ rests on a two-rung profile-normalisation (**prof**) ladder in the X sector: a weight- ≤ 14 instance (48 MB, solver 443 s, replay 62 s) gives $d_X \geq 15$, and Lemma P (all X -logicals even) lifts it to $d_X \geq 16$; an exact-weight-16 instance (310 MB, solver 2,083 s, replay 1,587 s) excludes weight 16, and Lemma P again lifts it to $d_X \geq 18$. With the weight-18 witness for $d_X \leq 18$ and the shipped duality certificate for $d = d_X$, this gives $d = 18$; the totals shown are over both rungs. The audited symmetry-broken $d_X \geq 10, 12, 14$ ladder (a different encoding) is retained and independently corroborates $d_X \geq 12$. See §6 for the caveats. [§]Lower end $d_X \geq 16$ certified by the same method (**prof** $K = 14$ plus Lemma P; a $K = 12$ rung, 14.7 MB, corroborates $d_X \geq 14$); the passage to $d \geq 16$ uses the ZX-duality lemma, verified in $\mathbb{F}_2$ arithmetic for this code but — unlike $n = 288$ — not shipped as a standalone certificate. Upper end $d \leq 24$ is Bravyi et al.’s Table 3 value, cited and not certified here.

We do *not* claim to be first to a machine-checked distance for this code. The value $d = 12$ is Bravyi et al.’s [BCGMRY, Table 3], confirmed exactly at MIP gap zero by Cruz-Benito, Cross, Kremer and Faro [CKKF26] and reproduced by SAT in [CJL26]. The idea of a kernel-checked quantum distance proof is LEAN-QEC’s [ELWT26], and their public repository reports the gross code completed; see §1.4 for the dated record. What is ours is the certificate format, the symmetry-free variant, and the trusted base.

(B) $[[288, 12, 18]]$, and $[[360, 12, \leq 24]]$. We determine $d([288, 12, 18]) = 18$ exactly. The upper bound is a weight-18 witness; the lower bound $d_X \geq 18$ is a two-rung ladder in a profile-normalisation (**prof**) encoding — a weight- ≤ 14 instance and an exact-weight-16 instance — each lifted across the intervening odd weight by Lemma P, with the duality certificate carrying

d_X to d . The value is not in dispute and we are not disputing it: Bravyi et al. assert $d = 18$ for this code exactly, by integer programming, and their Table 3 marks it without the “ $\leq$ ” that flags their upper-bound-only entries (Remark 6.1). What they do not supply — and what nobody supplied before — is a checkable artifact; our contribution is to *confirm* their value with one, not to change it. The strongest lower bound previously *reported* by a method capable in principle of certifying was $d \geq 11$, from Chen, Jafari and Lai [CJL26] at a 7200s timeout, solver-asserted, with no proof files in their repository and with every configuration running out of time. As far as the sweep of §1.4 could determine, ours is the first machine-checkable determination of this distance on record. The same construction gives the first lower bound of any kind for $[[360, 12, \leq 24]]$, whose exact distance is unknown: we certify $16 \leq d \leq 24$, the lower end by the identical method and the upper end cited from [BCGMRY]. Section 6 states the one genuine dependency plainly — the weight-16 exclusion is proved only in the **prof** encoding, corroborated by an independent encoding to $d_X \geq 12$ but not at the top of the ladder.

(C) The rest of the family. The values $d = 6, 10, 10$ for $[[72, 12, 6]]$, $[[90, 8, 10]]$, $[[108, 8, 10]]$, and the distances of Steane, Golay, the rotated surface codes and the five-qubit code, are all prior art; several are computed exactly, without certificates, in [CJL26]. For Steane, Golay, $[[72, 12, 6]]$ and $[[90, 8, 10]]$, replayable proof artifacts also exist in the LEAN-QEC repository, so “certified” is shared ground there and our contribution for those four is the trusted base, not primacy. We include them because a certificate format is worth nothing without a regression suite, and because six of them admit an independent brute-force cross-check (§7.2).

1.4. Provenance and credit. This subsection was written after a first-hand check of the prior art on August 4, 2026, in the course of which two claims in an earlier internal draft of these results turned out to be wrong. What follows is the corrected record. Table 2 is the claim-by-claim version.

The codes. The bivariate-bicycle family, including $[[144, 12, 12]]$, is Bravyi, Cross, Gambetta, Maslov, Rall and Yoder’s [BCGMRY]. All five parity-check matrices in our corpus were reconstructed from the construction in that paper and are byte-identical to it (§7.1). Their distances were computed there by the mixed-integer-programming method of Landahl, Anderson and Rice [LAR11]. Every distance value we certify for a BB code, and every distance value we certify for Steane, Golay, the surface codes and the five-qubit code, was already known. This note contributes *certificates for known numbers* — including the exact value $d = 18$ at $n = 288$, new only in the sense of being independently checkable — plus one genuinely new bound, $d \geq 16$ for $[[360, 12, \leq 24]]$, whose exact distance remains open. For $[[360, 12, \leq 24]]$ the parity-check matrices are rebuilt from the polynomial specification by `check_prof.py` and are not separately shipped.

Machine-checked quantum distance proofs. The approach is LEAN-QEC’s [ELWT26] (Ehatamm, Lee, Wu and Tao, 15 May 2026): a SAT encoding of the distance problem whose refutation is replayed inside the Lean 4 kernel. Their paper is candid about the gross code: it reports $[[144, 12, 12]]$ as dispatched by an external `cvc5` call rather than in the kernel, states that the 144-qubit instance pushes their certificate-replay budget beyond what they can verify in the kernel, and names kernel-side replay at that size as their next concrete engineering target. **Their repository has since moved past the paper.** At commit `c73827d` of `VerifiedQC/Lean-QEC`, dated 2026-07-10, a notes file reports that the $[[144, 12, 12]]$ BB code distance has been fully verified with `bv_decide` in about thirty minutes, by a pipeline whose final step translates the LRAT proof of unsatisfiability back into Lean and checks it with the

kernel. We therefore make *no* priority claim for a machine-checked gross-code distance, and we have removed such a claim wherever it appeared in our own working notes.

Several differences remain, and we state them as observations of a moving repository at a single commit, which may already be stale. At `c73827d`, the file `BB144.lean` contains two `sorry` placeholders — `BB144_X_ker_rank` at line 69 and `BB144_Z_ker_rank` at line 72 — through which the end-to-end theorem `BB144_dist_12` routes, so that theorem is not `sorry`-free as committed; three further lemmas use `native_decide`, which their own paper notes extends the trusted base with Lean’s compiler; and no LRAT artifact is committed for the 144-qubit instance, though LRATs are committed for smaller codes. Their 144-qubit encoding uses sorted-location symmetry constraints, i.e. it is symmetry-broken; we found no counterpart anywhere to the symmetry-free proofs of §5. We note also, since it would be easy to overstate their coverage in the other direction, that `BB108.lean` at the same commit carries `sorry` at lines 120 and 132 with its `bv_decide` invocation commented out, so their kernel-checked ladder should not be described as reaching $n = 108$ either. None of this is a criticism of work in progress; a repository ahead of its paper is a good problem to have. It is the reason we describe our contribution as a trusted base rather than a first, and we have made these statements as specific as we can so that they can be checked and, when they go stale, corrected.

Mathematical programming, independently. Cruz-Benito, Cross, Kremer and Faro [CCKF26] (1 June 2026, IBM) revisit the gross-code distance with a mixed-integer linear program and close it to a MIP gap of zero, confirming $d = 12$ exactly. That is a stronger form of the same kind of evidence as [BCGMRY]: an exhaustive search whose exhaustiveness is attested by the solver’s own optimality certificate rather than exported as a checkable object. We cite it as the current reference point for the value, and note that the gap it leaves — a MILP optimality proof is not a file — is precisely the one this note fills.

SAT-based distance computation at scale. Chen, Jafari and Lai [CJL26] (29 May 2026) compute BB distances with a battery of solvers, obtaining $d = 12$ for $[[144, 12, 12]]$ exactly and, for $[[288, 12, 18]]$, lower bounds of 11 under a 7200s budget. Their results are solver-asserted — their repository `guluchen/QDistSAT` contains no proof or certificate files — and their $d \geq 11$ is the strongest quantity previously published for that code *as a lower bound*, as against the exact value $d = 18$ asserted in [BCGMRY]. Our determination $d = 18$ improves on it and, for the first time, backs the distance of this code with a replayable artifact.

The even-weight lemma, and symmetry breaking. The ladder that reaches $d = 18$ uses two facts, neither of which is ours. Lemma P — that a CSS code all of whose H_Z -columns have odd weight has $1_n \in \text{rowsp}(H_Z)$ and hence an all-even Z -kernel — is elementary and is stated outright by Okada and Kasai [OK26] (Section V-A); we use it and claim it nowhere. The profile-normalisation reduction (Lemma S, §6) is an instance of lexicographic-leader symmetry breaking, a technique due to Crawford, Ginsberg, Luks and Roy [CGLR96] and automated in tools such as BreakID [DBBD16] and satsuma [And24]; exploiting the cyclic symmetry of quasi-cyclic codes in minimum-distance search is standard as well. What we constructed is one specific realisation of the *full* translation-orbit quotient for a bivariate-bicycle code inside a certifiable distance instance. We measure it against a lex-leader baseline over the same group (§6) but have not benchmarked it against those automated tools, which were not installable in our environment; we therefore make no novelty claim against them, and state the construction only at the strength that baseline supports.

Infrastructure. The LRAT format and the checking discipline are due to Cruz-Filipe, Heule, Hunt, Kaufmann and Schneider-Kamp [CFHKS17], building on `drat-trim` [HHW13]; our internal checker implements RUP-with-hints against that format. The solver is CaDiCaL [Bie20]. The cardinality encoding is Sinz’s sequential counter [Sin05]; the XOR gates are Tseitin’s

[Tse68]. The CSS structure is Calderbank–Shor [CS96] and Steane [Ste96]; the stabilizer formalism is Gottesman’s [Got97].

Adjacent work on certificate import. PBLearn [Sze26] imports VeriPB pseudo-Boolean certificates into Lean 4. It is not quantum, and it requires Lean, so it sits on the opposite side of design constraint (ii); but it is the closest prior art we found to the general project of making solver output externally checkable in a small trusted base, and we cite it as such.

1.5. What is not claimed. We do not claim any new distance *value* for a code whose exact distance was already known: $d([[288, 12, 18]]) = 18$ is Bravyi et al.’s, and we confirm it rather than discover it. The one piece of genuinely new distance information is the lower bound $d([[360, 12, \leq 24]]) \geq 16$, whose exact value remains open. We do not claim priority for machine-checked quantum distance proofs. We do not claim novelty for the two lemmas that close the $n = 288$ ladder: the even-weight Lemma P is Okada–Kasai’s, and the profile-normalisation Lemma S is an instance of standard lex-leader symmetry breaking; nor do we claim that the **prof** encoding improves on automated symmetry-breaking tools (BreakID, satsuma), which we did not run. We flag, rather than hide, that the top rung of the $n = 288$ ladder — the exclusion of a weight-16 logical, which is what separates $d \geq 18$ from $d \geq 16$ — is proved only in the **prof** encoding: an independent, differently structured encoding corroborates the ladder to $d_X \geq 12$ but not there, and the completeness of **prof** rests on the on-paper Lemma S (§4, §6). We do not certify k : the code dimensions in Table 1 are recomputed, not certified, though side condition (c) of Theorem 3.1 pins the logical dimension implicitly, and the audit recomputed every k independently. We do not claim the corpus is free of defects: four are reported in §8 — one fixed in this release, three not — including a genuine latent soundness hole in a checker branch that no shipped certificate exercises. And we do not claim novelty for the framing itself: this is a verification contribution, and the reason to read it is the audit of §7, not the numbers in Table 1.

2. CODES, DISTANCES, AND THE TWO CERTIFICATE PROBLEMS

We work over $\mathbb{F}_2$. A binary vector $x \in \mathbb{F}_2^n$ has weight $\text{wt}(x)$ and support $\text{supp}(x)$. For a matrix M , $\text{rowsp}(M)$ is its row space, and for a permutation π of $\{0, \dots, n-1\}$ we write $M\pi$ for the column-permuted matrix $(M\pi)_{r,i} = M_{r,\pi(i)}$.

Definition 2.1 (CSS code). A CSS code is a pair $H_X \in \mathbb{F}_2^{m_X \times n}$, $H_Z \in \mathbb{F}_2^{m_Z \times n}$ with $H_X H_Z^\top = 0$. Its dimension is $k = n - \text{rank } H_X - \text{rank } H_Z$. An X -logical is a vector x with $H_Z x = 0$; it is *nontrivial* if $x \notin \text{rowsp}(H_X)$. Set

$$d_X = \min\{\text{wt}(x) : H_Z x = 0, x \notin \text{rowsp}(H_X)\},$$

and d_Z symmetrically with X and Z interchanged. The code distance is $d = \min(d_X, d_Z)$.

The identity $d = \min(d_X, d_Z)$ is the standard CSS fact and we use it without further comment; it is item 5 in the trusted-base inventory of §4. For a general (non-CSS) stabilizer code we use the symplectic form ω on $\mathbb{F}_2^{2n}$, $\omega(p, q) = \sum_i (p_i q_{n+i} + p_{n+i} q_i)$, a stabilizer matrix $S \in \mathbb{F}_2^{m \times 2n}$ with pairwise-commuting rows, and qubit weight $\text{wt}_q(e) = \#\{i : e_i = 1 \text{ or } e_{n+i} = 1\}$.

Certifying d splits into two problems of completely different character. An *upper* bound is existential and needs only a witness. A *lower* bound is universal: it asserts that a search space of size $\binom{n}{\leq K}$ intersected with a coset is empty, and no small object witnesses that directly. The design question this note answers is what the smallest honest replayable object for the second problem looks like.

Item	Earliest source we verified	Status here
The BB codes; $d = 12$ for $[[144, 12, 12]]$; $d = 18$ for $[[288, 12, 18]]$; $d \leq 24$ for $[[360, 12, \leq 24]]$	Bravyi et al. [BCGMRY], Table 3 (MIP method of [LAR11])	Re-verified, not claimed (the $n = 288$ value now also certified checkably; the $n = 360$ upper bound cited)
$d = 12$ for $[[144, 12, 12]]$ confirmed exactly by MILP, MIP gap 0	Cruz-Benito, Cross, Kremer, Faro [CKKF26], 1 June 2026	Re-verified, not claimed; no certificate emitted there
$d_X = d_Z$ for BB codes	Bravyi et al. [BCGMRY], supplemental lemma	Cited; only the explicit permutation certificate is ours
Kernel-checked SAT distance proofs for quantum codes	LEAN-QEC [ELWT26], 15 May 2026	Not claimed
Machine-checked $[[144, 12, 12]]$ distance	LEAN-QEC repository, commit c73827d, 2026-07-10	Not claimed
d computed exactly by SAT for $n \leq 144$ BB codes	Chen–Jafari–Lai [CJL26], 29 May 2026	Re-verified, not claimed
$d \geq 11$ for $[[288, 12, 18]]$ by SAT	Chen–Jafari–Lai [CJL26]	Improved to the exact value $d = 18$, with replayable artifacts
Even weight of the Z -kernel (Lemma P), used to step the ladder by 2	Okada–Kasai [OK26], Sec. V-A	Cited, not claimed
Lex-leader symmetry-breaking predicates; cyclic-symmetry search	Crawford–Ginsberg–Luks–Roy [CGLR96]; BreakID [DBBD16]; satsuma [And24]	Cited; prof realises the same group’s full quotient, but is not benchmarked against these tools
<i>Offered as new:</i> The certificate format of §3 and its machine-checked exactness conditions; the symmetry-free $[[144, 12, 12]]$ proofs; the first machine-checkable determination on record that $d([[288, 12, 18]]) = 18$, and the first lower bound of any kind for $[[360, 12, \leq 24]]$ — both via the profile-normalisation ladder of §6, whose two supporting lemmas (the even-weight Lemma P and a standard symmetry break) are not ours; the ZX-duality permutation — Bravyi et al.’s lemma, not ours — packaged as a ~ 15 ms checkable certificate; and a trusted base of four standard-library Python readers, 1,128 lines, with no solver and no proof assistant in it.		

TABLE 2. Claim-by-claim provenance.

3. THE CERTIFICATE FORMATS

3.1. Upper bounds: witness pairs. An X -sector upper-bound certificate is a pair $(x, z) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$ together with a declared weight w . The checker (`check_witness.py`, 95 lines) verifies

$$H_Z x = 0, \quad H_X z = 0, \quad x \cdot z = 1, \quad \text{wt}(x) = w.$$

Soundness is immediate: $H_Z x = 0$ makes x an X -logical; every element of $\text{rowsp}(H_X)$ is orthogonal to z because $H_X z = 0$, so $x \cdot z = 1$ forces $x \notin \text{rowsp}(H_X)$. Hence $d_X \leq w$. Three matrix–vector products; no solver, no proof, no search. The gross-code witnesses are 970-byte files and check in 7 ms.

The general stabilizer form is the same shape with ω in place of the dot product: (e, f) with $\omega(s, e) = \omega(s, f) = 0$ for every stabilizer row s , and $\omega(e, f) = 1$, certifies $d \leq \text{wt}_q(e)$. The five-qubit code carries this variant end to end.

3.2. Lower bounds: a CNF the checker writes itself. Fix a sector, say X , and a bound K . The canonical CNF $\Phi_K(H_Z, P)$ over variables $x_1, \dots, x_n$ (plus auxiliaries) asserts

- (1) $H_Z x = 0$, one Tseitin XOR chain per row of H_Z , with the chain output forced false;
- (2) $b_j = z_j \cdot x$ for each row z_j of a *pairing* matrix P , one XOR chain each, output free;
- (3) the clause $(b_1 \vee \dots \vee b_p)$;
- (4) a Sinz sequential at-most- K counter [Sin05] on $x_1, \dots, x_n$;
- (5) optionally, forced unit literals (§3.3).

A lower-bound certificate is a JSON file naming H_X, H_Z, P , the value K , the sector, and for each instance a list of forced literals and an LRAT file. The checker `check_lower.py` (481 lines) does four things: verifies the side conditions of Theorem 3.1; *regenerates* $\Phi_K(H_Z, P)$ from the raw matrices in a fixed variable order; replays the LRAT against its own clause list; and, for symmetry-broken certificates, verifies the hypotheses of Lemma 3.2.

The shipped `.cnf` files are never read. They are informational, and the audit confirmed this by appending a contradictory clause pair to one of them and observing the check pass unchanged (§7).

Theorem 3.1 (Exactness of the CSS encoding). *Let $H_X \in \mathbb{F}_2^{m_X \times n}$, $H_Z \in \mathbb{F}_2^{m_Z \times n}$ and $P \in \mathbb{F}_2^{p \times n}$ satisfy*

- (a) $H_X H_Z^T = 0$;
- (b) $H_X P^T = 0$;
- (c) $\text{rank} \begin{pmatrix} H_Z \\ P \end{pmatrix} = n - \text{rank } H_X$.

Then

$$\{x \in \mathbb{F}_2^n : H_Z x = 0, Px \neq 0\} = \ker H_Z \setminus \text{rowsp}(H_X),$$

the set of nontrivial X -logicals. Consequently $\Phi_K(H_Z, P)$ is satisfiable if and only if there is a nontrivial X -logical of weight at most K ; an unsatisfiability proof for $\Phi_K(H_Z, P)$ certifies $d_X \geq K + 1$.

Proof. Put $V = \ker H_Z \cap (\text{rowsp } P)^\perp = (\text{rowsp } H_Z + \text{rowsp } P)^\perp$. By (a), $\text{rowsp}(H_X) \subseteq \ker H_Z$; by (b), $\text{rowsp}(H_X) \subseteq (\text{rowsp } P)^\perp$. Hence $\text{rowsp}(H_X) \subseteq V$. By (c),

$$\dim V = n - \text{rank} \begin{pmatrix} H_Z \\ P \end{pmatrix} = \text{rank } H_X = \dim \text{rowsp}(H_X),$$

so $V = \text{rowsp}(H_X)$. Now for $x \in \ker H_Z$ we have $Px \neq 0$ iff $x \notin (\text{rowsp } P)^\perp$ iff $x \notin V = \text{rowsp}(H_X)$, which is the claim. The second sentence follows because clauses (1)–(3) of Φ_K encode exactly $H_Z x = 0 \wedge Px \neq 0$ and clause group (4) encodes $\text{wt}(x) \leq K$. $\square$

Both directions of Theorem 3.1 are load-bearing and for different reasons. Condition (b) gives *soundness*: without it a satisfying assignment need not be a nontrivial logical, and UNSAT would prove nothing about the code. Condition (c) gives *completeness*: without it the encoding could miss logicals, and UNSAT would prove less than claimed. The checker verifies

(a) and (b) by explicit $\mathbb{F}_2$ inner products and (c) by two Gaussian eliminations over bitmask integers, on every certificate, before it looks at the proof. This is why the pairing matrix P — which the pipeline produced, and which a hostile pipeline could try to weaken — is not trusted input. Negative control 7 of §7 deletes one of its twelve rows and the checker rejects on condition (c).

3.3. Symmetry breaking, and the orbit lemma. For the largest instances the search space can be cut by the code’s own automorphisms. A symmetry-broken certificate ships permutations $\pi_1, \dots, \pi_r$, a block boundary b , and *two* instances with prescribed forced literals.

Lemma 3.2 (Orbit lemma). *Let $G \leq S_n$ be generated by permutations π such that*

- (i) π preserves the partition $\{0, \dots, b-1\} \sqcup \{b, \dots, n-1\}$;
- (ii) $\text{rowsp}(H_X \pi) = \text{rowsp}(H_X)$ and $\text{rowsp}(H_Z \pi) = \text{rowsp}(H_Z)$;

and suppose G acts transitively on each of the two blocks. If a nontrivial X -logical of weight $\leq K$ exists, then one exists whose support either contains qubit 0, or is disjoint from $\{0, \dots, b-1\}$ and contains qubit b .

Proof. By (ii), $x \mapsto x \circ \pi$ maps $\ker H_Z$ to itself and $\text{rowsp}(H_X)$ to itself, hence maps nontrivial X -logicals to nontrivial X -logicals, and it preserves weight. Let x be nontrivial of weight $\leq K$. If $\text{supp}(x)$ meets the left block, pick i in the intersection and $g \in G$ with $g(0) = i$, possible by transitivity; then $y = x \circ g$ has $y_0 = 1$. Otherwise $\text{supp}(x)$ meets the right block, say at $i \geq b$; pick $g \in G$ with $g(b) = i$; then $y = x \circ g$ has $y_b = 1$, and y still vanishes on the left block because g preserves blocks. $\square$

The two cases are exactly the two forced-literal patterns the checker demands: instance 0 has **forced** = $[1]$ and instance 1 has **forced** = $[-1, \dots, -b, b+1]$. Everything Lemma 3.2 hypothesizes is machine-checked: that each shipped array is a permutation, that it preserves the blocks (index comparison), that it is a rowspace automorphism of both H_X and H_Z (rank identity $\text{rank}(H) = \text{rank}(H \cup H\pi)$), that the generated group is block-transitive (breadth-first search over the generators and their inverses), and that the two forced patterns are literally the two above. Only the four-line implication in the proof is human-verified. For $[[144, 12, 12]]$ and $[[288, 12, 18]]$ the two shipped permutations were independently identified in the audit as the torus translations $x^1 y^0$ and $x^0 y^1$.

Remark 3.3. The gross code does not need this lemma. The 868 MB X proof and the 672 MB Z proof are single-instance certificates with empty **forced** lists, and the checker’s non-symmetry branch asserts exactly that: one instance, no forced literals. So $d([[144, 12, 12]]) = 12$ is certified with the orbit lemma removed from the trusted base. The symmetry-broken X certificate is retained because it is $7\times$ smaller and $17\times$ faster to replay, and because it demonstrates the mechanism at $n = 144$ before it is used in earnest at $n = 288$.

3.4. The ZX-duality certificate. For BB codes $d_X = d_Z$. This is not our observation: it is a lemma in the supplemental material of Bravyi et al. [BCGMRY], invoked there in exactly the way we invoke it — to halve the work — with the remark that the hypothesis in question “is satisfied for BB LDPC codes”. What we add is an artifact: an explicit permutation, shipped as a file, that turns the lemma into something a reader checks in about 15 ms instead of something a reader believes.

Lemma 3.4 (Duality). *Let Π be a permutation of the n qubits with $\text{rowsp}(H_X \Pi) = \text{rowsp}(H_Z)$ and $\text{rowsp}(H_Z \Pi) = \text{rowsp}(H_X)$. Then $x \mapsto y$, $y_i = x_{\Pi(i)}$, is a weight-preserving bijection from nontrivial X -logicals to nontrivial Z -logicals, and $d_X = d_Z$.*

Proof. Applying Π^{-1} to the two hypotheses gives $\text{rowsp}(H_X) = \text{rowsp}(H_Z\Pi^{-1})$ and $\text{rowsp}(H_Z) = \text{rowsp}(H_X\Pi^{-1})$. Let h be a row of H_X . Then $h \cdot y = \sum_i h_i x_{\Pi(i)} = \sum_j h_{\Pi^{-1}(j)} x_j = (h \circ \Pi^{-1}) \cdot x$, and $h \circ \Pi^{-1} \in \text{rowsp}(H_X\Pi^{-1}) = \text{rowsp}(H_Z)$, so $h \cdot y = 0$ whenever $H_Z x = 0$. Thus $H_X y = 0$. If $y \in \text{rowsp}(H_Z)$, say $y = cH_Z$, then $x_j = y_{\Pi^{-1}(j)}$ exhibits $x \in \text{rowsp}(H_Z\Pi^{-1}) = \text{rowsp}(H_X)$, contradicting nontriviality of x . Weight is preserved because Π permutes coordinates. The same argument with X and Z interchanged gives the inverse map, so the correspondence is a bijection and $d_X = d_Z$. $\square$

The certificate is the permutation Π ; the checker (`check_duality.py`, 73 lines) does two rank comparisons. The corpus ships one for each of the five BB codes, and each checks in a few milliseconds to a few tens of milliseconds — 3.8, 4.7, 6.1, 9.9 and 37.9ms for $n = 72, 90, 108, 144, 288$ on a loaded machine. In every case Π is the composite of the block swap with $(a, b) \mapsto (-a, -b)$ on the torus indices.

Remark 3.5 (The $n = 288$ certificate, and when it appeared). The audit of §7 found no duality certificate for $[[288, 12, 18]]$ (discrepancy D2 of §8): the generator listed the code but had not been re-run, so what the corpus certified there was $14 \leq d_X \leq 18$ rather than $14 \leq d \leq 18$. The auditor independently confirmed that the same permutation family does satisfy both rowspace identities at $n = 288$, so the mathematics was never in doubt — only the artifact was missing, which is exactly the condition this paper exists to avoid. The generator has since been re-run and `bb288/duality.json` now ships. It is the one certificate in the release that is *not* among the audit’s 47 checks and not among the 182 hashes in `manifest.json`, both of which predate it; it passes `check_duality.py`, and its permutation was verified a second time, from scratch, against independently loaded matrices. With it — and, in the August 6 extension, with the profile-normalisation ladder of §6 that raises d_X from 14 to 18 — $d = 18$ is certified rather than merely asserted, and Table 1 is worded accordingly. No Z -sector witness is needed: $d = \min(d_X, d_Z) = d_X$ once duality is in hand.

3.5. The symplectic variant. For a general stabilizer code the encoding uses $3n$ variables — u_i, v_i for the X - and Z -components and q_i for “qubit i is in the support”, tied by $(\neg u_i \vee q_i), (\neg v_i \vee q_i), (u_i \vee v_i \vee \neg q_i)$ — with XOR chains for the symplectic products and the at-most- K counter on the q ’s.

Proposition 3.6. *Let $S \in \mathbb{F}_2^{m \times 2n}$ have pairwise ω -orthogonal rows, let $L \in \mathbb{F}_2^{\ell \times 2n}$ satisfy $\omega(s, \lambda) = 0$ for all rows s of S and λ of L , and suppose $\text{rank} \begin{pmatrix} S \\ L \end{pmatrix} = \text{rank } S + \ell = n + k$ where $k = n - \text{rank } S$. Then $\{e : \omega(s, e) = 0 \ \forall s, \exists \lambda \ \omega(e, \lambda) = 1\}$ is exactly the set of nontrivial logical operators.*

Proof. Let C be the ω -centralizer of $\text{rowsp}(S)$, of dimension $2n - \text{rank } S = n + k$. The radical of $\omega|_C$ is $\text{rowsp}(S)$. The rows of S and L all lie in C and span a subspace of dimension $\text{rank } S + \ell = n + k$, so they span C . If $e \in C$ and $\omega(e, \lambda) = 0$ for every row of L , then $\omega(e, \cdot)$ vanishes on C , so e lies in the radical, i.e. $e \in \text{rowsp}(S)$. The converse inclusion is immediate. $\square$

The five-qubit code $[[5, 1, 3]]$ carries this path end to end: a 128-byte witness and a 1,514-byte LRAT for $d \geq 3$. It exists to keep the non-CSS branch of the checker exercised; see also defect D4 in §8, which is a missing guard in exactly that branch.

4. THE TRUSTED BASE

A certificate is only as good as the list of things its reader must still believe. Here is that list, split into what is re-verified per certificate by code one can read (CHECKED) and what is argued in prose and verified nowhere (ASSUMED).

4.1. Software. Four files, 1,128 lines of Python in total: `check_witness.py` (95), `check_duality.py` (73), `check_lower.py` (481), and `check_prof.py` (479). The first three, 649 lines, are the checkers the independent audit of §7 read and exercised; the fourth was added with the August 6 extension to verify the profile-normalisation certificates behind $d = 18$ at $n = 288$ and the $n = 360$ lower bound, and shares no code with the other three. Their only imports are `gzip`, `json`, `os`, `subprocess`, `sys`, `tempfile` and `time`. No `numpy`, no compiled helper, no network. CPython and the operating system must be trusted. If the optional `--external` path is used to delegate LRAT replay to a compiled checker, that binary and the temporary-file marshalling re-enter the trusted base; the audit of §7 never used it, so for that audit the trusted base contained no compiled code at all. `check_prof.py` is, if anything, stricter than `check_lower.py`: it rebuilds H_X, H_Z from the code’s polynomial specification rather than reading `HX.txt/HZ.txt`, and it requires the shipped CNF to match its own regeneration clause-for-clause before replaying the proof against the regenerated clauses.

Not trusted, and demonstrably so: CaDiCaL; the generating pipeline (`certify.py`, `qec_lib.py`, `gen_duality.py`, `manifest.py`); the shipped `.cnf` files; `meta.json`; `manifest.json`; and the prose of this paper. All of them can be deleted and every certificate still verifies.

Remark 4.1 (The checker moved after the audit). The auditor of §7 read a 419-line `check_lower.py`; the file shipped with this release is 481 lines. The difference is one addition: an optional truncated Bailleux–Boufkhad totalizer as an alternative to the Sinz cardinality encoding, added during exploratory work on this family. It is selected only by a certificate that declares `"cardinality": "totalizer"`, and *no `check_lower` certificate in this release declares it*: all take the Sinz default, which is the code path the audit read and exercised. A reader minimising trusted-base surface can delete the totalizer branch and re-run everything. We flag the drift rather than quietly re-using the audit’s line count, because the size of that number is one of the claims. To close the loop, the entire audited corpus was re-run against the shipped 481-line checker before release — 20 witnesses, 5 duality certificates and 23 lower-bound certificates, 48 in all, every one accepted, including a second pure-Python replay of the 2.94 GB proof. Those re-runs were made on a machine under heavy load and their wall-clock times are correspondingly three to five times the audit’s; the pass/fail outcomes are what they establish. The August 6 profile-normalisation certificates behind $d = 18$ at $n = 288$ and the $n = 360$ bound are the province of a separate checker, `check_prof.py` (§4, §6), and postdate both the audit and this re-run; each was replayed independently before release.

4.2. Machine-checked, per certificate.

- CSS orthogonality $H_X H_Z^T = 0$ — condition (a).
- Every pairing row in the kernel of the quotient matrix — condition (b), soundness.
- The rank/spanning identity — condition (c), completeness.
- For the symplectic branch: commutativity of S, L in the centralizer, and the two rank identities of Proposition 3.6.

- For `_sym` certificates: that each permutation is a permutation, preserves blocks, is a rowspace automorphism of both H_X and H_Z ; that the generated group is block-transitive; and that the forced patterns are exactly the two Lemma 3.2 licenses.
- For duality certificates: both rowspace identities.
- Regeneration of the entire CNF from the raw matrices, and RUP replay of the LRAT against *that* clause list.

4.3. Assumed.

- (1) *The cardinality encoding is complete.* For the `check_lower` certificates this is Sinz’s sequential at-most- K counter [Sin05]: every weight- $\leq K$ assignment extends to a satisfying assignment of the counter variables. If it were accidentally over-constraining, UNSAT would mean less than claimed; the emitted clause set was read against [Sin05] and is the standard one. The `check_prof` certificates behind $d = 18$ at $n = 288$ and the $n = 360$ bound instead use a Bailleux–Boufkhad totalizer (with per-row exact counters and a lexicographic-leader predicate, §6.1); its completeness, and that of the lex-leader, is assumed on the same footing.
- (2) *The Tseitin XOR gate is definitional* [Tse68]: the four clauses per gate encode $u = a \oplus b$ and are always extendable.
- (3) *The orbit lemma*, Lemma 3.2 — four lines, proved above, machine-checked hypotheses, human-verified implication. Used by the symmetry-broken $[[144, 12, 12]]$ X certificate and by the retained symmetry-broken $[[288, 12, 18]]$ ladder (the $d_X \geq 10, 12, 14$ rungs that now corroborate the main result). **Not used** by the symmetry-free $[[144, 12, 12]]$ certificates (Remark 3.3), nor by the profile-normalisation rungs, which use Lemma S instead.
- (4) *The duality lemma*, Lemma 3.4 — likewise, checked hypotheses, human-verified implication. It carries d_X to d at $n = 288$ (shipped `duality.json`, replayed by `check_duality.py`) and at $n = 360$ (verified by rowspace comparison in $\mathbb{F}_2$ arithmetic, but not shipped as a standalone certificate for that code).
- (5) *The even-weight lemma*, Lemma P (§6): if $1_n \in \text{rowsp}(H_Z)$ then every $x \in \ker H_Z$ has even weight, so d_X is even. Elementary and credited to Okada–Kasai [OK26]; its hypothesis is machine-checked by `check_prof.py`, which exhibits the combiner c with $c^\top H_Z = 1_n$ ($|c| = 72$ at $n = 288$, $|c| = 90$ at $n = 360$). It is what lets an even- K UNSAT certify $d_X \geq K + 2$, and it is used at both $n = 288$ prof rungs and both $n = 360$ rungs.
- (6) *The profile-normalisation lemma*, Lemma S (§6): adding a row-profile invariant and a one-slice lex-leader to the weight-bounded search leaves satisfiability unchanged. Its hypothesis — that the two generating translations are automorphisms of both H_X and H_Z — is machine-checked by `check_prof.py` (rowspace equality). The lemma’s conclusion is on paper; it is tested against brute force on 31 small codes and, at $n = 288$, by re-normalising the known weight-18 logical, but it is *not* cross-checked by a second independent encoding at the decisive weight-16 rung. This is the one soundness dependency peculiar to the $d = 18$ result; §6 states it in full.
- (7) *The CSS fact* $d = \min(d_X, d_Z)$, used to turn a certified pair into a statement about the code. No script touches it.
- (8) *LRAT semantics.* The internal checker implements RUP-with-hints: negate the lemma, unit-propagate through the hinted clauses in order, demand a conflict. It skips hints that are already satisfied or non-unit — sound, since skipping can only fail to find a conflict, never invent one — and it *refuses* negative (RAT) hints outright, which

incidentally establishes that every proof in the corpus is pure RUP. It requires an empty clause to be derived and verified.

- (9) *That the matrices are the intended code.* No certificate establishes this; the checkers take `HX.txt` and `HZ.txt` at face value. Section 7.1 closes this for the five BB codes at byte level.

In one sentence: a skeptic must believe that four short standard-library Python files do what they appear to do, that CPython and the OS are not lying, that the cardinality (Sinz for `check_lower`, a totalizer for `check_prof`) and Tseitin encodings are standard and complete, that the four-line duality lemma and the four-line orbit lemma are correct, that $d = \min(d_X, d_Z)$, and — for the $d = 18$ determination at $n = 288$ and the $n = 360$ lower bound alone — the elementary even-weight lemma and the profile-normalisation lemma. Everything else can be thrown away.

5. THE GROSS CODE

$[[144, 12, 12]]$ is the $\ell = 12$, $m = 6$ member of the BB family, with $A = x^3 + y + y^2$ and $B = y^3 + x + x^2$, $H_X = [A \mid B]$, $H_Z = [B^\top \mid A^\top]$ [BCGMRY]. Its certified distance decomposes as follows; all byte counts are the shipped files.

statement	artifact	bytes	solver (s)	replay
$d_X \leq 12$	<code>witness_X.json</code>	970	0.025	7.4 ms
$d_Z \leq 12$	<code>witness_Z.json</code>	970	0.029	7.3 ms
$d_X \geq 12$, symmetry-broken	<code>lower_X_K11_sym.json</code> +2 LRATs	123,857,070	45.0	10.1 s
$d_Z \geq 12$, symmetry-free	<code>lower_Z_K11.json</code> +1 LRAT	671,988,205	227.3	72.9 s
$d_X \geq 12$, symmetry-free	<code>lower_X_K11.json</code> +1 LRAT	867,803,294	342.3	176.4 s
$d_X = d_Z$	<code>duality.json</code>	103 + 466	—	9.9 ms

Total solver time for the complete symmetry-free route is 569.6s: under ten minutes on a laptop. Total replay time for that route, in pure Python with no compiled code anywhere, is 249.3s. The two symmetry-free proofs are what let us say that $d([[144, 12, 12]]) = 12$ holds on the trusted base of §4 *minus* item 3.

The value $d = 12$ is not new; it is [BCGMRY, Table 3], obtained by mixed integer programming [LAR11], confirmed exactly at MIP gap zero by [CCKF26], and reproduced by SAT in [CJL26]. A machine-checked proof of it is not new either: the LEAN-QEC repository reports one as of 2026-07-10 (§1.4). What is offered here is the shape of the artifact — an 868 MB file plus a 481-line reader, with no symmetry hypothesis, no solver and no proof assistant — and the fact that it replays inside 51 MB of memory.

Remark 5.1 (Timing variability). The replay times above are the audit’s measurements on an otherwise idle machine. A re-run of the symmetry-broken X certificate performed while writing this note, under load, took 30.4s rather than 10.1s. All timing figures in this paper should be read with a factor of three of slack; the byte counts and the pass/fail outcomes are exact.

6. $[[288, 12, 18]]$: THE EXACT DISTANCE, AND $[[360, 12, \leq 24]]$

For the $\ell = m = 12$ member, $A = x^3 + y^2 + y^7$, $B = y^3 + x + x^2$, we determine the distance exactly:

$$d([[288, 12, 18]]) = 18,$$

certified end to end. The upper bound $d_X \leq 18$ is the shipped weight-18 witness; duality (§3.4) gives $d = d_X$. The content is the lower bound $d_X \geq 18$, reached by two rungs of an encoding different from the rest of this note, closed by an elementary parity lemma.

Remark 6.1 (What the literature says, stated correctly). Bravyi et al. assert $d = 18$ for this code exactly. An earlier internal draft of our results claimed instead that they screened it with a heuristic and conceded $d \leq 18$ “is unlikely to be tight”; that was a misreading of their circuit-level distance $d_{\text{circ}} \leq 18$, a different quantity, and it is corrected here. Table 3 of [BCGMRY] lists $[[288, 12, 18]]$ with no “ $\leq$ ”, while $[[360, 12, \leq 24]]$ and $[[756, 16, \leq 34]]$ carry one, and the caption states that “ $\leq d$ ” marks upper-bound-only entries; the supplemental material states that the actual distance “of each candidate code was computed using the integer linear programming method”. So $d = 18$ is their value, obtained by ILP and without a checkable artifact. What this section adds is the artifact: the first determination of $d([[288, 12, 18]]) = 18$, as far as our sweep could tell, that a third party can replay. It confirms their value; it does not change it.

6.1. Two lemmas. Lemma P (even weight) [Okada–Kasai [OK26], Sec. V-A]. If $1_n \in \text{rowsp}(H_Z)$, pick c with $c^\top H_Z = 1_n^\top$; then for every $x \in \ker H_Z$, $1_n \cdot x = c^\top H_Z x = 0$ over $\mathbb{F}_2$, so $\text{wt}(x)$ is even and d_X is even. The hypothesis holds for every BB code here — each H_Z column has weight 3, so $c = 1_{\ell m}$ works — and `check_prof.py` exhibits the combiner ($|c| = 72$ at $n = 288$). Lemma P is elementary and is not ours; it turns an even- K UNSAT into $d_X \geq K + 2$.

Lemma S (profile normalisation). Index qubits by (b, r, s) with $b \in \{0, 1\}$, $r \in \mathbb{Z}_\ell$, $s \in \mathbb{Z}_m$, and let $T = \mathbb{Z}_\ell \times \mathbb{Z}_m$ act by translation. Write the row profile $w_r(x) = |\text{supp}(x) \cap \{(b, r, \cdot) : b\}|$ and the row-0 pattern $P_0(x) = (x_{0,0,\cdot}, x_{1,0,\cdot})$. Then every nonzero T -orbit contains an x whose profile $(w_0, \dots, w_{\ell-1})$ is lexicographically maximal among its ℓ rotations and whose row-0 pattern is lexicographically maximal among its m rotations. (Translation by (a, c) rotates the profile by a independently of c , which fixes a ; then c ranges over the m rotations of the row-0 pattern. This is a lexicographic-leader construction in the sense of Crawford, Ginsberg, Luks and Roy [CGLR96], arranged so that the profile half rides the cardinality counter the instance already needs.) Because T is an automorphism group of the code and preserves weight, restricting the weight-bounded search to such x leaves satisfiability unchanged, realising the full $|T| = \ell m$ translation quotient.

The `prof` encoding adds exactly these two conditions to the canonical CNF $\Phi_K(H_Z, P)$ of §3.2. `check_prof.py` machine-checks the *hypothesis* of Lemma S — that the generators $(1, 0)$ and $(0, 1)$ are rowspace automorphisms of both H_X and H_Z — and, as with every certificate here, regenerates the whole CNF from the polynomial spec and replays the LRAT against it. What it does *not* do is prove the lemma’s conclusion: that is on paper, exactly as the orbit and duality lemmas of §3.3–§3.4 are. Its support is (a) agreement with brute force on 31 small BB codes, across the modes `none`, `anchor2`, `prof` and `prof+exact` for every K up to $d_X + 2$, with zero disagreements; and (b) at $n = 288$, a direct control — the known weight-18 logical, re-verified to be genuine, has a T -translate satisfying the two conditions (profile $[8, 0, 0, 4, 0, 0, 4, 0, 0, 2, 0, 0]$), and the exact-weight-18 `prof` instance with that translate pinned is satisfiable, so the reduction does not delete a true minimum-weight logical.

6.2. The ladder to 18. Each rung is a single `prof` instance in the X sector, checked by `check_prof.py` (standard library only, CNF regenerated from the spec). Rung 1, a weight- ≤ 14 instance (4,939 vars, 27,101 clauses; 48MB gzipped proof, solver 443s, pure-Python replay 62s, 447,281 lemmas), is UNSAT: no nontrivial X -logical has weight ≤ 14 , so $d_X \geq 15$, and Lemma P lifts this to $d_X \geq 16$. Rung 2, an *exact*-weight-16 instance (5,249 vars, 31,208

clauses; 310 MB gzipped proof, solver 2,083 s, replay 1,587 s, 2,335,793 lemmas), is UNSAT: no nontrivial X -logical has weight exactly 16, and Lemma P kills weight 17, so $d_X \geq 18$. The weight-18 witness gives $d_X \leq 18$ and duality gives $d = 18$. Total: 2,526 s of solver and 358 MB of proof, checked in about 28 minutes of pure Python. There is no $K = 17$ rung: Lemma P makes the exact-weight 16 instance the only thing left to refute, which is why the encoding does not have to reach the $K = 17$ ladder step the earlier draft projected out of reach.

6.3. What the top rung rests on, exactly. Three things should be said without euphemism.

First, *the passage from $d_X \geq 16$ to $d_X \geq 18$ is single-encoding*. Only rung 2 — the weight-16 exclusion — separates $d \geq 18$ from $d \geq 16$, and it exists only in the **prof** encoding. The corpus also ships a *different* encoding for this code: the symmetry-broken Sinz/anchor ladder of the August 4 release, whose audited rungs reach $d_X \geq 14$ (the 2.94 GB proof), and a freshly re-run independent totalizer/Sinz cross-check confirms $d_X \geq 12$ from the polynomial spec. So the ladder is corroborated in a second, differently structured encoding up to $d_X \geq 14$; everything above that — the $d_X \geq 16$ prof rung and the $d_X \geq 18$ weight-16 exclusion — is at present carried by **prof** alone.

Second, *completeness of **prof** is Lemma S, an on-paper lemma*. Its hypothesis is machine-checked and its conclusion is tested as in §6.1, but it is not reduced to the exactness theorem the way the Sinz encoding of §3.2 is. A reader who declines to trust Lemma S still has $d_X \geq 14$ from the audited non-**prof** ladder; the steps to 16 and 18 are the ones that use Lemma S.

Third, *the encoding is characterised only relative to a baseline*. The $144\times$ translation group is the entire affine automorphism budget of this code — an automorphism search over the affine group finds nothing larger — and **prof** spends all of it. Measured against this repository’s own best encoding (**anchor2**+totalizer) on the same laptop, **prof** takes $5\text{--}11\times$ fewer CaDiCaL conflicts across $[[144, 12, 12]]$ and $[[288, 12, 18]]$; measured against a lex-leader over the *same* group (the partial lex-leader an off-the-shelf tool emits for this symmetry), it takes about $3.8\text{--}4\times$ fewer conflicts at comparable or smaller formula size — at $n = 288$, $K = 11$, for instance, 23,369 clauses and 36,465 conflicts for **prof** against 47,348 clauses and 147,263 conflicts for the lex-leader. We did not run BreakID [DBBD16] or satsuma [And24]; whether either would match **prof**’s cheap realisation of this quotient is untested, and no novelty is claimed against them.

For context on the value of the artifact: Chen, Jafari and Lai [CJL26] run a battery of solver configurations on this code under 7200 s timeouts, all of which time out with $d \geq 11$ the best lower bound reached and no proof files in their repository. The two rungs above settle the strictly harder exact question on one laptop, with proofs any reader can replay.

6.4. $[[360, 12, \leq 24]]$: a first lower bound. The same construction transfers to the $\ell = 30$, $m = 6$ member, $A = x^9 + y + y^2$, $B = y^3 + x^{25} + x^{26}$, which is off the $\ell = m$ diagonal; the profile is therefore placed on the larger cyclic factor, the axis being a certificate field the checker reads. Its parameters reproduce $n = 360$, $k = 12$. Bravyi et al. [BCGMRY] give only an upper bound, listing it as $[[360, 12, \leq 24]]$; no lower bound of any kind has been reported, and Chen–Jafari–Lai time out on it. A **prof** $K = 14$ instance (14,015 vars, 79,397 clauses; 43 MB gzipped, solver 231 s, replay 40 s, 428,498 lemmas) certifies that no nontrivial X -logical has weight ≤ 14 , so $d_X \geq 15$, and Lemma P (here $|c| = 90$) raises it to $d_X \geq 16$; a $K = 12$ rung (14.7 MB) corroborates $d_X \geq 14$. The ZX-duality permutation $(b, r, s) \mapsto (1 - b, -r, -s)$ was verified for this code by the same rowpace comparison the checker uses — it exchanges $\text{rowsp}(H_X)$ and $\text{rowsp}(H_Z)$ — so $d_Z = d_X$ and

$$16 \leq d([360, 12, \leq 24]) \leq 24,$$

the lower end certified for the first time and the upper end cited from [BCGMRY]. Two limits are stated at point of use. First, what a shipped `prof` certificate replays directly for this code is $d_X \geq 16$; unlike $n = 288$, no standalone `check_duality` certificate ships for $n = 360$, so the passage from $d_X \geq 16$ to $d \geq 16$ rests on the duality lemma verified in exact $\mathbb{F}_2$ arithmetic (both rowspace identities hold) rather than on a one-command replayable artifact. Second, this bound rests on the same on-paper Lemmas P and S as the $n = 288$ result, and on no independent-encoding cross-check. It is offered at exactly that strength.

7. INDEPENDENT VERIFICATION

The corpus was re-checked by a separate agent instance with no access to the pipeline and no shared code, on the same machine, using `/usr/bin/python3` (CPython 3.9.6). Its report is `INDEPENDENT-VERIFICATION.md`. Summary: 47 checks run, 47 passed, 0 failed; 5,459,315,046 bytes (5.08 GiB) of LRAT replayed in approximately 746 s of wall clock; peak resident set 79 MB; all 182 SHA-256 entries in `manifest.json` re-hashed and matching. Because `lrat-check` is not present in the repository (defect D3), *every* proof — including the 2.94 GB one — was replayed with the internal pure-Python checker. That is the stronger result. Eleven negative controls, built by the auditor on scratch copies, were all correctly rejected (§7.3), and six codes were cross-checked by brute force (§7.2). This audit, not the distance values, is the load-bearing part of the paper. The profile-normalisation certificates behind the exact $d = 18$ at $n = 288$ and the $n = 360$ lower bound (§6) postdate this audit and are outside its 47 checks; each was replayed independently by the standard-library `check_prof.py`, which rebuilds the matrices from the polynomial spec, but they did not pass through the separate auditor.

7.1. Are these the right matrices? This is the check the certificates cannot perform on themselves, and it was done first. The auditor rebuilt H_X and H_Z for all five BB codes directly from the construction in [BCGMRY] — $n = 2\ell m$, $x = S_\ell \otimes I_m$, $y = I_\ell \otimes S_m$ over $\mathbb{F}_2$ with S_r the cyclic shift, $H_X = [A \mid B]$, $H_Z = [B^\top \mid A^\top]$ — using that paper’s Table 3 parameters. For every one of `bb72`, `bb90`, `bb108`, `bb144`, `bb288`, the shipped `HX.txt` and `HZ.txt` are **byte-identical** to the reconstruction: no row operations, no column permutation, no block swap. Independently recomputed $k = n - \text{rank } H_X - \text{rank } H_Z$ gives 12, 8, 8, 12, 12, matching the published parameters, with row weight 6 and column weight 3 throughout. Every algebraic side condition was then re-run a second time, in the auditor’s own implementation, against the *reconstructed* matrices rather than the shipped ones; all agreed.

7.2. Brute force where brute force is possible. For the six codes small enough, the auditor computed the minimum distance by exhaustive Gray-code enumeration over a kernel basis, with code sharing nothing with either the pipeline or the checkers: Steane, surface $d = 3$, surface $d = 5$, Golay, the five-qubit code, and surface $d = 7$ (a 2^{25} search, 242 s). Six exact distances, six agreements with the certified values. From `[[72, 12, 6]]` upward the search space is 2^{42} and beyond, which is the entire point of the certificates.

7.3. Negative controls. Eleven tampered artifacts were built on scratch copies and all eleven were rejected, each with the specific error one would want: a flipped witness bit (“witness does not commute”), an understated weight, a relabelled K (“LRAT lemma 2015 NOT verified”), a deleted empty clause, a truncated proof, a live-but-wrong hint id, a gutted pairing matrix (rejected on condition (c), “ $31 \neq 42$ ”), a flipped bit of H_X (rejected on condition (a)), a swapped sector label, a corrupted hint id, and an identity duality permutation. In addition, a contradictory clause pair was appended to `bb72/lower_X_K5.cnf` and the check passed

unchanged and in the same time — a positive control on the claim that the shipped CNF is genuinely never read.

8. DEFECTS FOUND

Reported verbatim, because a paper about trusted bases that suppresses its own audit findings is not one.

D1 (documentation). The results file cites a `tamper_test/` directory that does not exist in the repository. The claim is true — the controls were rebuilt from scratch and all rejected (§7.3) — but the citation is dangling.

D2 (real, touched a headline, now closed). At audit time there was no duality certificate and no Z -witness for $[[288, 12, 18]]$, so the corpus certified $14 \leq d_X \leq 18$, not $14 \leq d \leq 18$: the generator listed that code and had evidently not been re-run. The underlying mathematics was verified independently by the auditor, so the claim was true and uncertified, which is precisely the condition this paper exists to avoid. This is the one defect of the four that is fixed in this release: the generator has been re-run, `bb288/duality.json` ships, it passes `check_duality.py`, and its permutation has been re-verified from scratch. The caveats are that this certificate post-dates the audit and is therefore outside its 47 checks, and that `manifest.json` — whose 182 entries the audit re-hashed — has deliberately not been regenerated, so it does not list the new file. See Remark 3.5.

D3 (documentation). `run_all.sh` requires `tools-drat-trim/lrat-check`, which is absent with no source or URL, so the `lrar-check` timings previously quoted are not reproducible from the repository as shipped. The pure-Python path is reproducible and is what Table 1 reports. Two earlier pure-Python figures were also stale in the conservative direction (21s quoted versus 6.2s measured for $[[108, 8, 10]]$; 17.3s versus 10.1s for the symmetry-broken $[[144, 12, 12]]$ certificate), and the quoted throughput of 3.4 MB/s is closer to 12 MB/s.

D4 (latent soundness hole). In the CSS branch, `check_lower.py` requires a certificate with no symmetry block to have exactly one instance with an empty forced list. The symplectic branch has *no such guard*: it passes the certificate’s forced literals straight into the encoder, which emits them as unit clauses. The auditor demonstrated this by adding `"forced": [-1, -2, -3]` to the five-qubit certificate and obtaining `OK lower bound: d >= 3` from a proof that now covers only assignments avoiding those qubits. No shipped certificate is affected — the five-qubit certificate’s forced list is empty, and this was checked — but the checker would accept a forged one. The one-line fix is to mirror the CSS assertion. Two lesser observations in the same vein: the symmetry branch does not assert that there are exactly two instances (harmless, since each is replayed independently), and the CSS branch does not validate the sector label (a typo silently selects the other branch, whose side conditions then fail, as negative control 9 shows).

D4 is the interesting one. It is exactly the class of defect that the certificate discipline is supposed to make findable: a hole in 481 lines of readable Python, found by reading them, rather than a hole in a solver.

9. RELATION TO OTHER WORK

Heuristic upper bounds. QDistRnd [PSKK22] produces upper bounds with no performance guarantee; Stim’s logical-error search [Gid21] is documented as heuristic. Both are excellent for what they are. Our upper bounds differ only in shipping the witness.

Exact methods without artifacts. The mixed-integer-programming method of Landahl, Anderson and Rice [LAR11] is what [BCGMRY] used and remains the reference for the BB

distances; Cruz-Benito, Cross, Kremer and Faro [CCKF26] re-close the gross-code MILP to gap zero. Chen, Jafari and Lai [CJL26] push SAT and MaxSAT to this family at scale and explicitly do not log proofs. Webster, Jacob and Higgott [WJH26] survey the field and distinguish exact from heuristic methods carefully; their desiderata do not include exportable certificates, and an exhaustive search of their text turns up a single occurrence of certification, referring to internal optimality rather than to an artifact. None of these produces something a third party can replay. That gap, not the numbers, is what this note addresses.

Symmetry breaking, and even weight. The profile-normalisation reduction that reaches $d = 18$ (§6) is a lexicographic-leader symmetry break, the technique of Crawford, Ginsberg, Luks and Roy [CGLR96]; automated symmetry-breaking-predicate generators such as BreakID [DBBD16] and satsuma [And24] produce such predicates for a solver’s input, and exploiting the cyclic symmetry of quasi-cyclic codes in minimum-distance search is standard. We compare our construction against a lex-leader over the same translation group, not against those tools, which we could not install; the comparison and the resulting refusal to claim novelty against them are in §6.3. The even-weight lemma (Lemma P) is Okada and Kasai’s [OK26].

Proof-assistant approaches. LEAN-QEC [ELWT26] is the closest work and the one we have most to learn from: their location-indexed encoding is better than ours, and we did not need it only because we stopped at $n = 288$. The design difference is where the trust sits. Their artifact is replayable inside the Lean 4 kernel, which is a stronger guarantee than ours about the *checker*, at the cost of requiring Lean and Mathlib — a toolchain fetch — to obtain it. Ours is replayable by any CPython, which is a weaker guarantee about the checker and a much smaller barrier to a skeptic. Neither dominates. PBLearn [Sze26] imports VeriPB certificates into Lean 4 and is the nearest general prior art for the certificate-import problem.

What we could not find. A search of the public record on 2026-08-04 — arXiv, code search for LRAT together with quantum distance, and the issue trackers of Stim, **panqec**, **ldpc** and **qLDPC** — turned up no standalone quantum distance certificate requiring neither a solver nor a proof assistant to check. A negative cannot be proved this way; we record the scope of the search rather than a claim. The August 6 extension repeated the exercise for its own material: a full-text pass over the distance and symmetry-breaking sources cited here turned up no exportable-certificate distance proof at $n = 288$ or $n = 360$, and no earlier instance of the specific full-translation-orbit profile encoding, but it did not include a benchmark against automated symmetry-breaking tools, so the encoding’s standing against them is left open (§6.3).

10. REPRODUCIBILITY

The artifacts sit beside this note in **qec/**. To re-check anything:

```
# check_prof.py reads its LRAT gzipped: do NOT gunzip the
# *_prof_*.lrat.gz files. The check_lower proofs below are shipped
# uncompressed or regenerated (REGENERATE.md), so this block needs
# no gunzip step; for the wider corpus, gunzip the *non-prof*
# .lrat.gz as REGENERATE.md describes.

# (i) Runs on a fresh clone, from the shipped artifacts alone:
python3 check_witness.py certificates/bb144/witness_X.json
python3 check_duality.py certificates/bb144/duality.json
python3 check_prof.py   certificates/bb288/bb288_prof_K14.json # d_X>=16
python3 check_prof.py   certificates/bb360/bb360_prof_K12.json # >=14
python3 check_prof.py   certificates/bb360/bb360_prof_K14.json # d_X>=16
```



```
python3 check_duality.py certificates/bb288/duality.json          # d=d_X

# (ii) Each needs one large proof regenerated first, none in git
#      (REGENERATE.md items 1, 2, 4, 5):
python3 check_lower.py certificates/bb144/lower_X_K11.json      # 868MB
python3 check_lower.py certificates/bb144/lower_Z_K11.json      # 672MB
python3 check_lower.py certificates/bb288/lower_X_K13_sym.json  # 2.94GB
python3 check_prof.py  certificates/bb288/bb288_prof_K16_exact.json # =>d=18
```

No virtual environment, no packages, no compiled binary. Any CPython 3.8 or later should do; the audit used the macOS system interpreter, 3.9.6. Expect 176s, 73s and 414s respectively for the three `check_lower` large replays on an idle M4, more under load; the two $n = 288$ `prof` rungs replay in about 62s and 1,587s and the $n = 360$ rung in about 40s. `manifest.json` carries the SHA-256 and byte count of the 182 audited artifacts and `manifest.py` re-checks them; the profile-normalisation certificates postdate the manifest and are checked directly by `check_prof.py`, which regenerates each CNF from the polynomial spec. (The four commands in group (ii) — the two symmetry-free gross-code proofs, the $K=13$ rung at $n = 288$, and the exact-weight-16 `prof` instance — each require regenerating one large proof first, per `REGENERATE.md` items 1, 2, 4 and 5; on a fresh clone group (i) runs as shown and group (ii) does not. The expected replay figures below are for the already-regenerated proofs.)

What the public repository can and cannot hold. Five of the proofs — the two symmetry-free $[[144, 12, 12]]$ certificates, the first instance of each of the $K = 11$ and $K = 13$ rungs at $n = 288$, and the 310MB exact-weight-16 `prof` instance at $n = 288$ — are between 79MB and 646MB compressed and are not carried in git. Everything else is: the descriptors, the parity-check matrices, the pairing and permutation files, the witnesses, the duality certificates, the CNF inputs, the manifest, and every proof small enough to ship, up to and including the 30MB symmetry-broken $[[144, 12, 12]]$ certificate and the $K = 9$ rung at $n = 288$. So a reader who clones and runs nothing but CPython can still replay a certified $d = 3$ for all four sanity-tier codes, $d = 7$ for Golay and the $d=7$ surface code, $d = 6$ for $[[72, 12, 6]]$, $d = 10$ for $[[90, 8, 10]]$ and $[[108, 8, 10]]$, $d = 12$ for the gross code — by the symmetry-broken X certificate together with the duality certificate and the weight-12 witness, the two symmetry-free proofs being exactly the ones that do not fit — and $d_X \geq 10$ at $n = 288$. The August 6 extension adds, and ships, the profile-normalisation proofs for $d_X \geq 16$ at $n = 288$ (48MB) and for $d_X \geq 16$ at $n = 360$ (43 and 14.7MB), each replayable from a clone by `check_prof.py`. At $n = 288$ the shipped `duality.json` converts d_X to d ; at $n = 360$ the ZX-duality permutation is verified by the same rowspace comparison (§6.4) but is *not* shipped as a standalone `check_duality` certificate, so what a clone replays directly at $n = 360$ is $d_X \geq 16$, with $16 \leq d \leq 24$ following from that duality lemma. The one proof the extension does *not* ship is the 310MB exact-weight-16 instance that lifts $n = 288$ from $d_X \geq 16$ to $d_X \geq 18$: it exceeds the per-file limit, so a reader regenerates it from the shipped CNF — which `check_prof.py` reconstructs from the polynomial spec in any case — and then checks it. So a fresh clone replays $d_X \geq 16$ at both $n = 288$ and $n = 360$ directly, reaches $16 \leq d \leq 24$ at $n = 360$ through the (unshipped-for-360) duality lemma, and reaches the final $d = 18$ at $n = 288$ after one solver run on the shipped CNF. For the five omitted proofs the release ships `REGENERATE.md`: the exact `CaDiCaL` invocation for each, with the expected byte count and the SHA-256 the result must have. Regenerating them puts a solver back in the loop for the *production* of the proof, which is where a solver has always been allowed to sit; the check that follows still does not trust it. The generating pipeline (`certify.py`, `qec_lib.py`, `run_all.sh`) is included for completeness

and is not needed to check anything; it requires `numpy` and a `CaDiCaL` binary, and as noted in D3 its full `run_all.sh` path additionally expects an `lrat-check` binary that is not vendored.

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Finding that a result one has produced was produced first by someone else is the ordinary condition of working in a field that is moving faster than the writing; we have tried to record it accurately rather than minimally.

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